

Synchronization - Part 2

Operating Systems – EDA093/DIT401

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What to read (main textbook)

- Chapter 2.3.7-2.3.9, 2.3.11, 6.3, 2.4.1-2.4.6, 6.1-6.2, 6.5-6.6, 6.7.3-6.7.4

Agenda

- Bounded-buffer producer/consumer
- Resource allocation, deadlocks, and necessary conditions for deadlocks
- Dining philosophers
 - Breaking the circular wait condition
 - Breaking the no-preemption condition
 - Breaking the hold-and-wait condition
- Lamport's bakery algorithm
- Readers/Writers problem
- OS as arbitrator for deadlock avoidance/recovery

Agenda

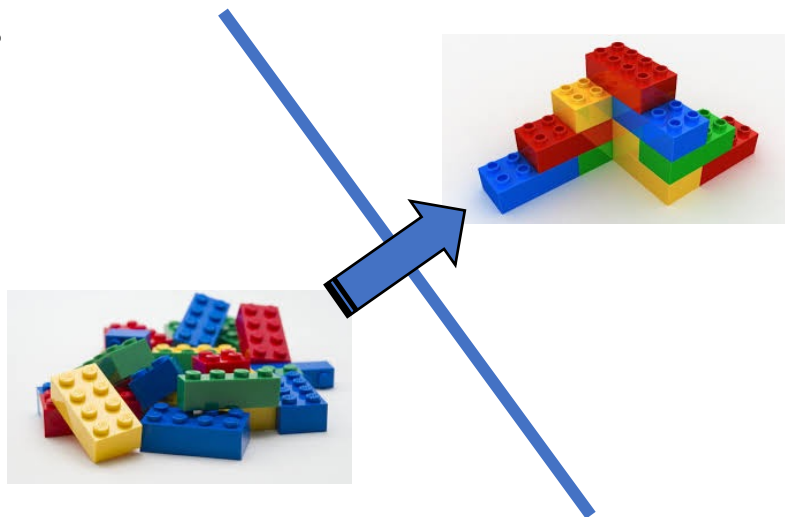
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The Bounded buffer producer-consumer problem

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By using primitives ...

- R/W variables
- RMW variables
- Semaphores, etc.
- ...

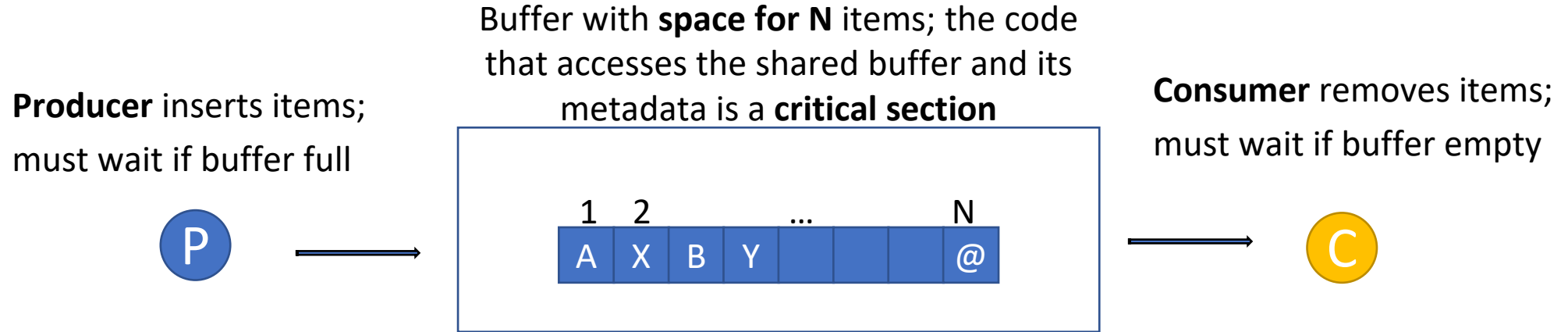


... we want to construct objects / solve specific synchronization problems

- 2-thread CS, n-thread CS
- Producer-consumer (bounded buffer)
- Dining philosophers
- Transactions
- ...

the Bounded buffer producer-consumer problem

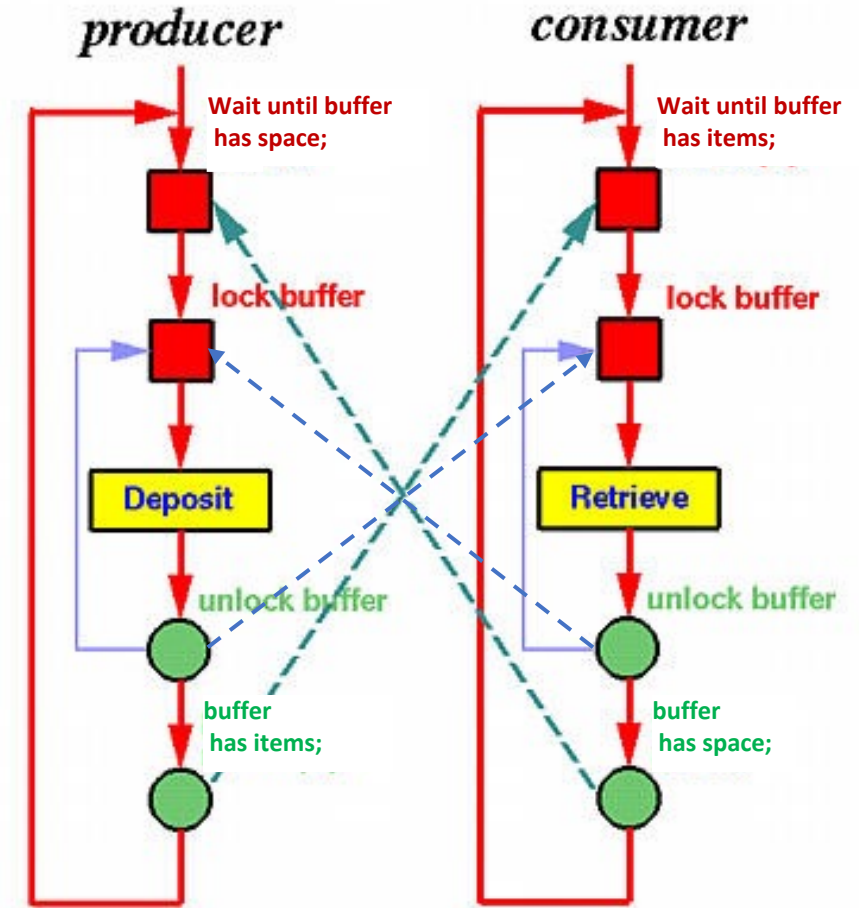
6



Solve this synchronization problem using semaphores

What synchronization do we need?

- **Producer** inserts items; must wait if buffer full
- **Consumer** removes items; must wait if buffer empty
- Accessing the buffer is a **critical section**



a solution to the Bounded buffer producer-consumer problem 8

Synchronization variables:

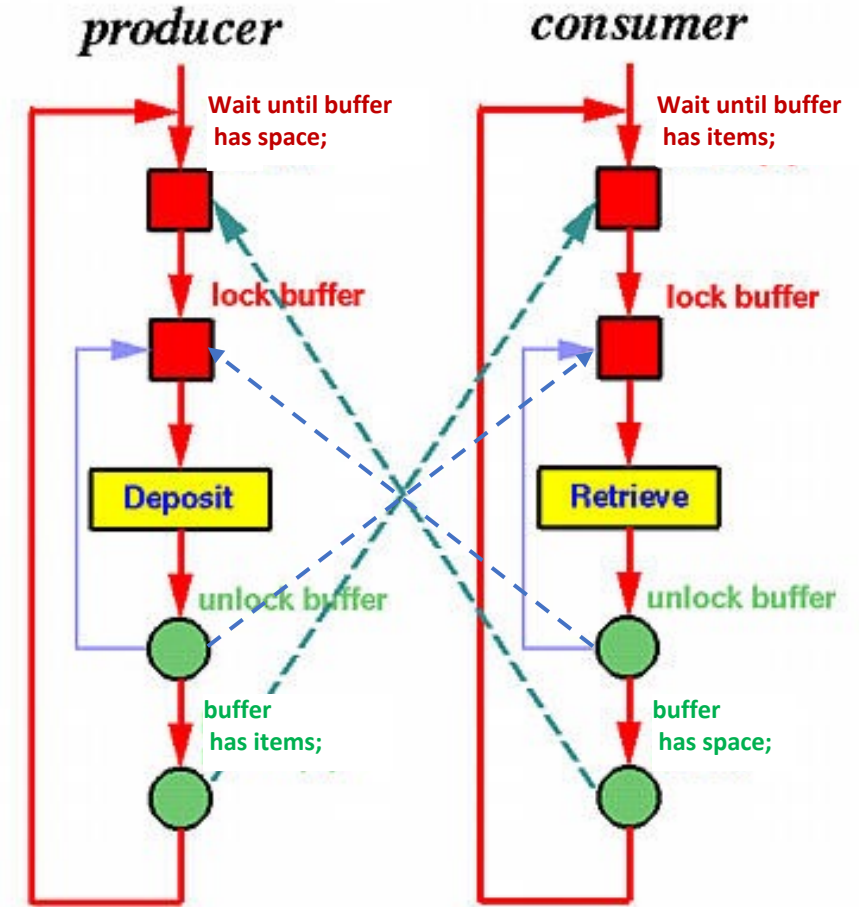
- Binary semaphore **mutex_sem** initialized to 1
- Counting semaphore **buffer-has-items** initialized to 0
- Counting semaphore **buffer-has-space** initialized to N

producer

```
do {  
    // produce item  
    wait(buffer-has-space);  
    wait(mutex_sem);  
    // add item to buffer  
    signal(mutex_sem);  
    signal(buffer-has-items);  
} while (TRUE);
```

consumer

```
do {  
    wait(buffer-has-items);  
    wait(mutex_sem);  
    // remove item from buffer  
    signal(mutex_sem);  
    signal(buffer-has-space);  
    // use the item  
} while (TRUE);
```



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Resource allocation

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- Processes/threads need resources (e.g., memory pages, a printer, access to parts of a shared data structure, etc.)
 - Our focus: reusable resources
- a human analogy: process = go fishing; needed resources: boat, fishing rod
- Mutual exclusion is needed for resources that cannot be shared



Structure of process/thread P

Repeat

Request resources (entry section)

critical section

Release resources (exit section)

remainder section

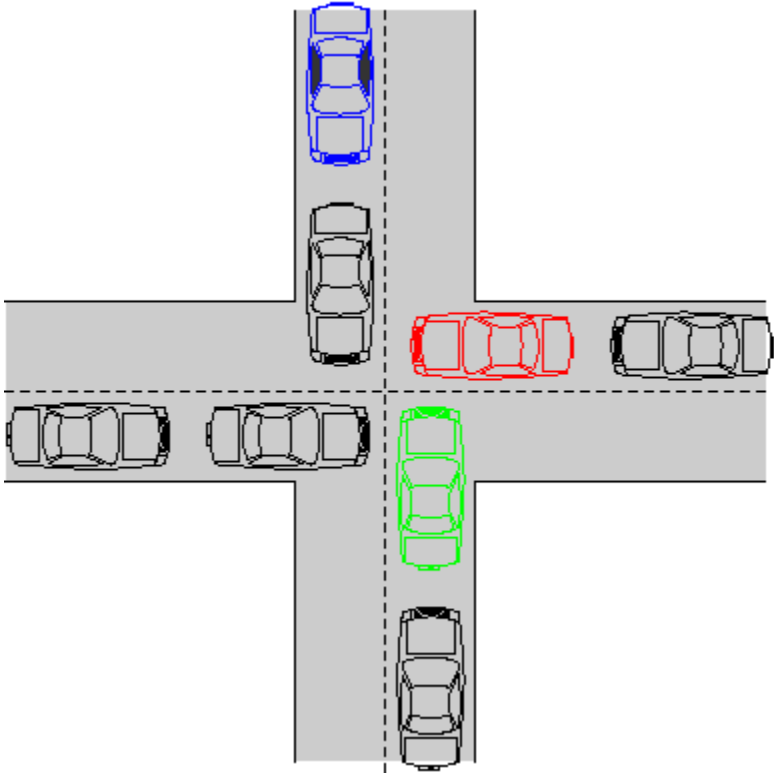
Forever

Problem formulation:

A solution for acquiring/releasing all the needed resources must **provide** the **entry** and **exit** sections **ensuring**:

- 1. Mutual Exclusion.**
- 2. Progress: no deadlock**
- 3. Fairness (e.g., Bounded Waiting, no starvation)**

What is a deadlock?

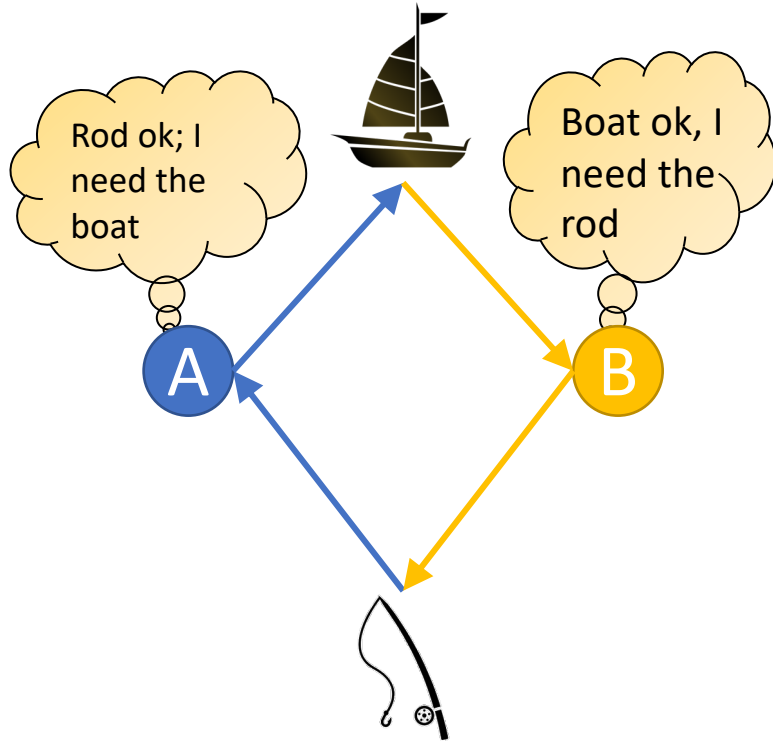


A set of processes/threads blocking each other so that none of them can proceed:

How can it occur?

4 necessary conditions for Deadlock [Coffman et al., 1971]

Theorem: all 4 conditions hold simultaneously when a deadlock occurs:

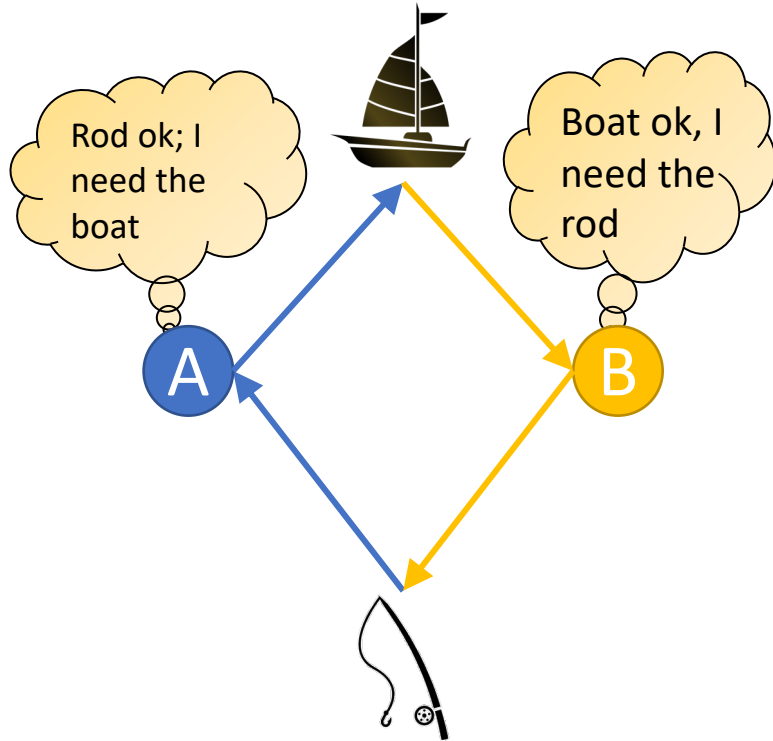


1. **Mutual exclusion:** only one process at a time can use a resource (that is not shareable).
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes.
3. **No preemption:** a resource need not be held until the entire task is complete; it is released voluntarily after the process finishes using it.
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next process in the chain

NOTE: we are talking about preemption of resources, this is not the process scheduling preemption

4 necessary conditions for Deadlock [Coffman et al 1971]

Theorem: all 4 conditions hold simultaneously when a deadlock occurs:



1. **Mutual exclusion:** only one process at a time can use a resource (that is not shareable).
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes. → **GET ALL RESOURCES AT ONCE**
3. **No preemption:** a resource can only be released by the process holding it, after that process has completed its task. → **THREADS RELEASE RESOURCES IF THEY MANAGE TO GET SOME BUT NOT ALL**
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next process in the chain. → **ACQUIRE RESOURCES IN A CERTAIN ORDER**

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Dining philosophers [Dijkstra, 1965]

n philosophers (processes); each philosopher P_i , when hungry, needs both left & right chopsticks, in order to eat

Structure of process/thread P

Repeat

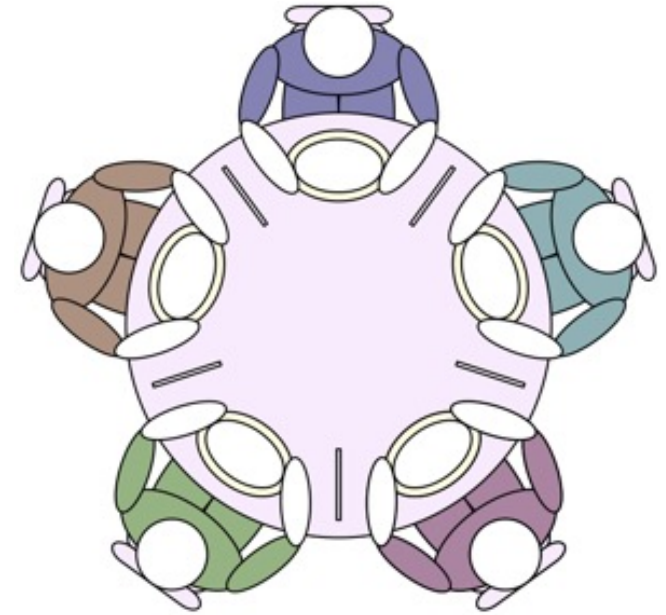
Request resources (left + right chopstick)

eat

Release resources (left + right chopstick)

think

Forever



Dining philosophers: pick-left-then-right approach

```
1.  Shared var c[0..n-1]: binary semaphore // one for
    each chopstick; initially all set to 1
2.  Pi:
3.  do
4.      wait c[i]; // pick left chopstick
5.      wait c[(i+1) mod n]; // pick right chopstick
6.      // Eat
7.      signal c[i]; // leave left chopstick
8.      signal c[(i+1) mod n]; // leave right chopstick
9.      // Think
10. forever
```

Recall the requirements:

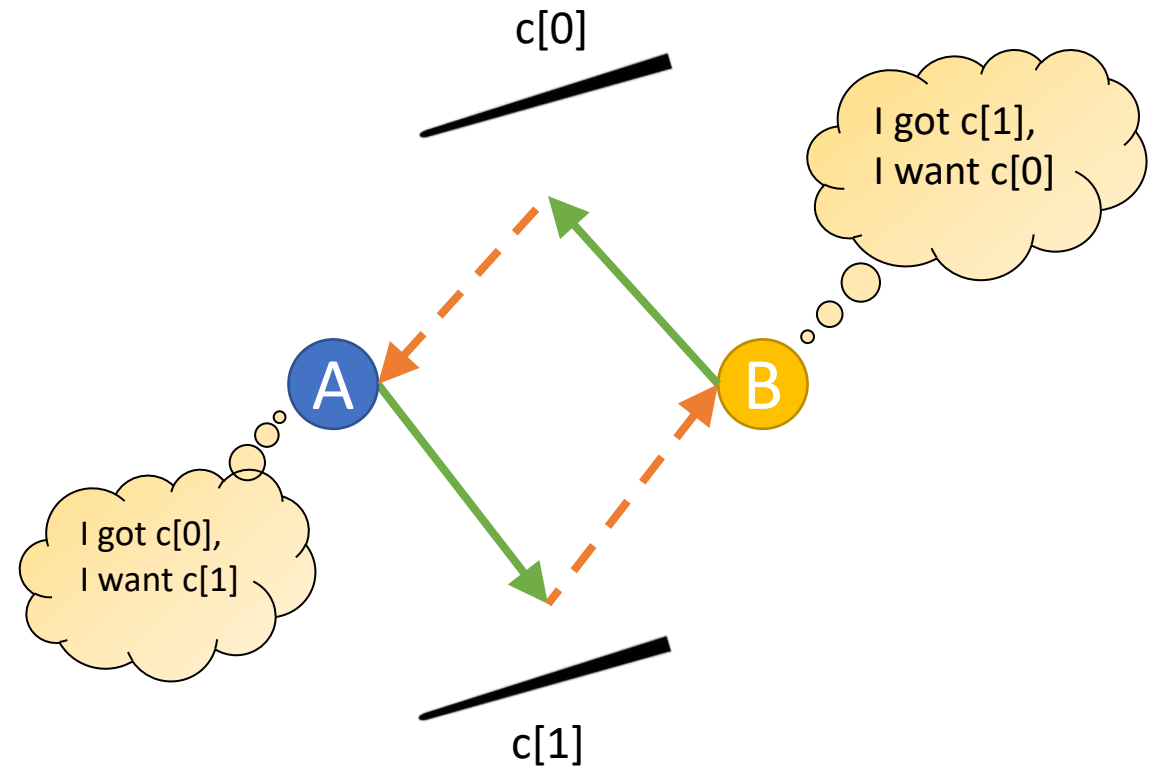
1. **Mutual exclusion:** each resource is used by only one process at a time
2. **Progress:** no deadlock
3. **Fairness:** FCFS, or no starvation, or other fairness formulation

Dining philosophers: pick-left-then-right approach

```
1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2. Pi:
3. do Hold and wait
4.   Wait c[i]; // pick left chopstick
5.   Wait c[(i+1) mod n]; // pick right chopstick
6.   // Eat
7.   Signal c[i]; // leave left chopstick
8.   Signal c[(i+1) mod n]; // leave right chopstick
9.   // Think No preemption
10. forever
```

Dining philosophers: pick-left-then-right approach

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   each chopstick; init all 1
2. Pi:
3. do                                     Hold and wait
4.   Wait c[i]; // pick left chopstick
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7.   Signal c[i]; // leave left chopstick
8.   Signal c[(i+1) mod n]; // leave right chopstick
9.   // Think                             No preemption
10. forever
```



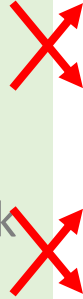
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Dining philosophers: pick-one-at-a-time without circular wait

```

1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2.  $P_i$  (except  $P_{n-1}$ ):
3. do
4.     Wait c[i]; // pick left chopstick
5.     Wait c[(i+1) mod n]; // pick right chopstick
6.     // Eat
7.     Signal c[(i+1) mod n]; // leave right chopstick
8.     Signal c[i]; // leave left chopstick
9.     // Think
10. forever
  
```



```

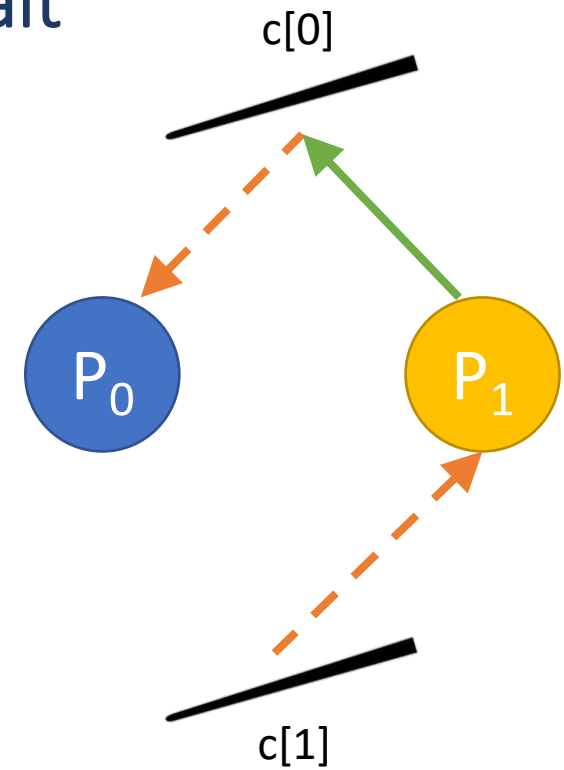
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   each chopstick; init all 1
2.  $P_{n-1}$ :
3. do
4.     Wait c[(i+1) mod n]; // pick right chopstick
5.     Wait c[i]; // pick left chopstick
6.     // Eat
7.     Signal c[i]; // leave left chopstick
8.     Signal c[(i+1) mod n]; // leave right chopstick
9.     // Think
10. forever
  
```

Dining philosophers: pick-one-at-a-time without circular wait

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```
1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pi (except Pn-1):
3.  do
4.    Wait c[i]; // pick left chopstick
5.    Wait c[(i+1) mod n]; // pick right chopstick
6.    // Eat
7.    Signal c[(i+1) mod n]; leave right chopstick
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9.    // Think
10. forever
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1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pn-1:
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6.    // Eat
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9.    // Think
10. forever
```



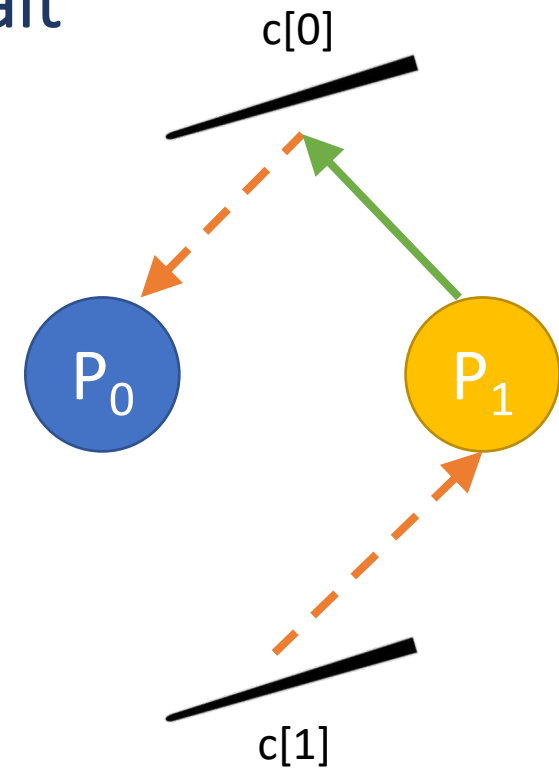
Dining philosophers: pick-one-at-a-time without circular wait

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8.    Signal c[i]; // leave left chopstick
9.    // Think
10. forever
  
```

```

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    for each chopstick; init all 1
2.  Pn-1:
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9.    // Think
10. forever
  
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Dining philosophers: break the no-preemption condition

```
shared var c[0..n-1]: of type chopstick_struct { // one for each chopstick
  s: bin-semaphore // init 1
  available: boolean //init true
}
```

```
Pi:
  local var holding_both_chopsticks: boolean; // initially false
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

```
trylock(c: chopstick_struct):
  wait(c.s)
  if c.available then {
    c.available := false
    ret:= true }
  else
    ret:= false;
  signal(c.s)
  return(ret)
```

Idea: when the second resource is not available, release the first one and retry



```
lock(c : chopstick_struct):
  repeat
  until (trylock(c))
```

```
release(c : chopstick_struct):
  wait(c.s)
  c.available := true
  signal(c.s)
```


Dining philosophers: break the no-preemption condition

- Mutual exclusion: ok
- Progress? no deadlock ... but livelock is possible!
- Fairness? a process can starve...

```
Pi:
  local var holding_both_chopsticks: boolean;
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

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Dining philosophers: Break the hold-and-wait condition



Idea: philosophers agree on who eats next instead of who gets a chopstick

```
shared var semaphore S[0 .. n-1] // init all 0
shared var semaphore mutex // init 1
shared var state[0 .. n-1] in {HUNGRY, THINKING, EATING} // all initialized as THINKING
Pi:
do
  // think
  enterCS(i) // i.e., get both chopsticks
  // eat
  exitCS(i) // i.e., leave chopsticks
forever
```

```
enterCS(i)
  wait(mutex)
  state[i] := HUNGRY
  help(i)
  signal(mutex)
  wait(S[i])
```

```
help(k)
  if state[k] == HUNGRY &&
    state[(k-1) mod n] != EATING &&
    state[(k+1) mod n] != EATING
  then {state[k] := EATING ; signal(S[k]) }
```

```
exitCS(i)
  wait(mutex)
  state[i] := THINKING
  help((i-1) mod n)
  help((i+1) mod n)
  signal(mutex)
```

Dining philosophers: Break the hold-and-wait condition

- Mutual exclusion: ok
 - Progress: no deadlock
 - Fairness: a process can starve...
-
- If you are interested in a solution with fairness too:
<https://dl.acm.org/doi/pdf/10.1145/62546.62567>

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Lamport's bakery algorithm

Critical section for n threads:

Idea: before entering its critical section, each thread gets a number. Holder of the smallest number enters the critical section.



The decentralized scheme generates numbers in non-decreasing order; e.g., 1, 2, 3, 3, 4, 5, 5, 5, 6 (note: tickets return to zero, and a later round of threads trying to enter the critical section may start again with tickets with value 1)

If threads P_i and P_j choose the same number:

if $i < j$, then P_i goes first; else P_j goes first.

I.e., we use threads' ids to break ties

Lamport's bakery algorithm

```

Shared var choosing: array [0..n - 1] of boolean (init false);
          number: array [0..n - 1] of integer (init 0);

repeat
  choosing[i] := true;
  number[i] := max(number[0], number[1], ..., number [n - 1])+1;
  choosing[i] := false;
  for j := 0 to n - 1 do begin
    while choosing[j] do [nothing]; // spin
    while number[j] ≠ 0 and (number[j],j) < (number[i], i) do [nothing]; // spin
  end;
  // critical section
  number[i] := 0;
  // remainder section
until false;

```

Why does it satisfy the 3 conditions:

Mutual exclusion: (no 2 threads A and B in CS concurrently): Consider the time between A's decision step and A's entry to CS; A decided to move because:

- B had higher number: when B checks , it will wait for A since A has smaller number
- or B was not interested; when B gets interested, it will choose a number > A's number, hence it will wait

Progress: the thread with the smaller (number, thread ID) can proceed

Fairness: If A waits for B and B exits and wants to enter CS again, if A still waits, B will choose a number > A's, number, hence B cannot bypass A

*This is a more **decentralized** method than e.g., Peterson's: no variable is "writable" by all threads*

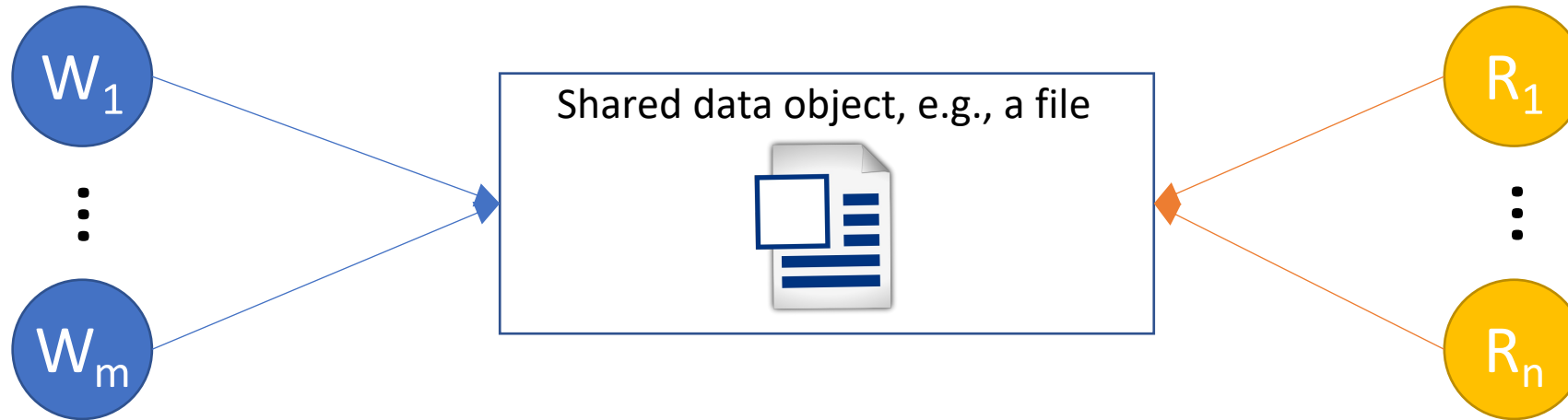
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Readers/Writers problem

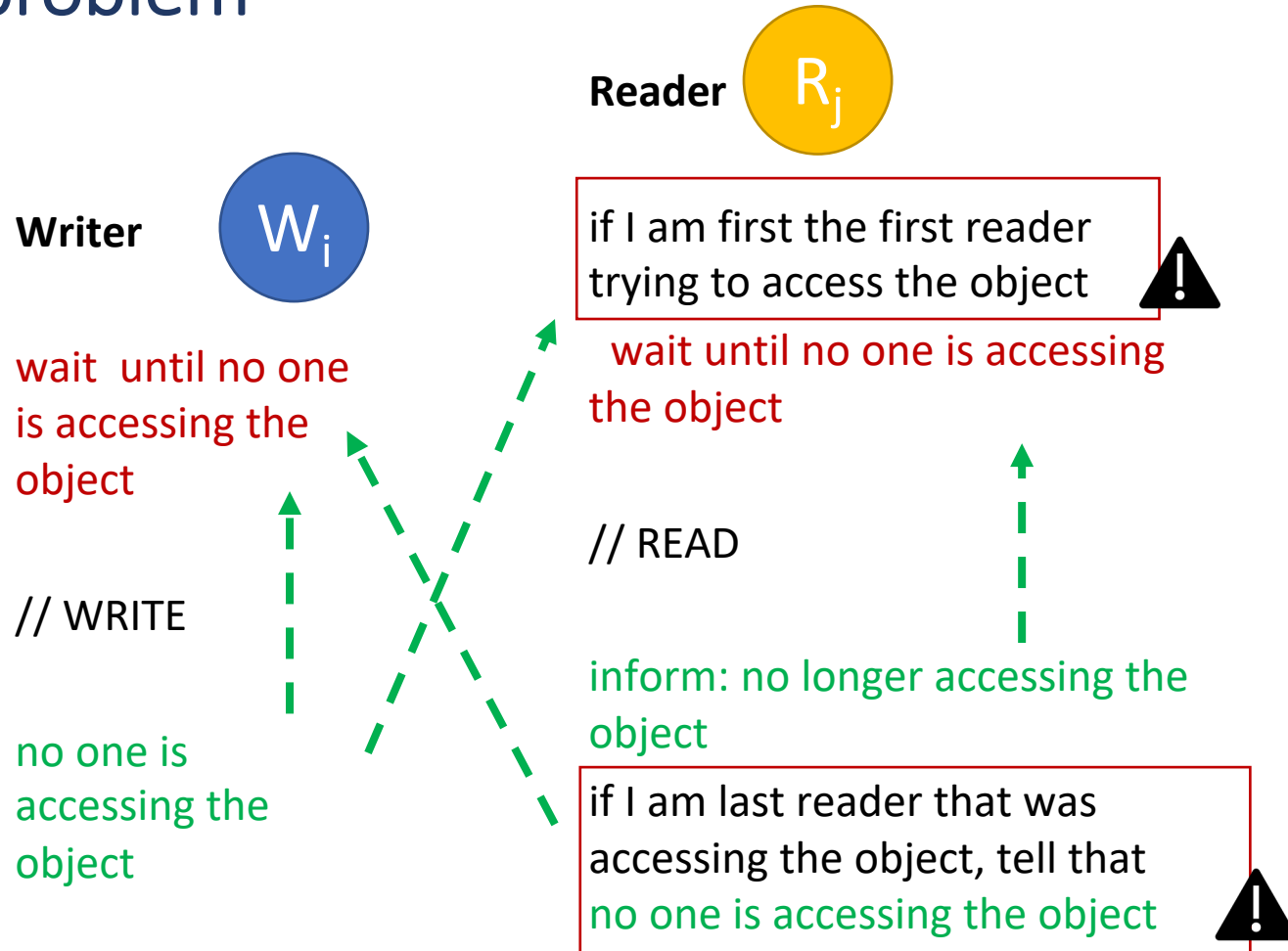
Writers write; must wait if some writer or reader is accessing the shared data object (i.e., *only one writer is allowed to write at a time*)

Readers read; must wait if some writer is accessing the shared data object (i.e., *multiple readers are allowed to read at a time but not concurrently with any writer*)



Solve this synchronization problem using semaphores, ensuring the presented safety properties and progress

Readers/Writers problem



Readers/Writers problem

shared var:

noone_accesses, protect_check: binary semaphore; // initially 1
rc: int ; // active readers counter, init 0



Writer



Repeat

```
wait(noone_accesses);
// WRITE
signal(noone_accesses)
forever
```

Reader



repeat

```
wait(protect_check); // CS to change and check rc variable
rc++;
if rc == 1 then wait(noone_accesses) // "first" reader: block writers or wait
if some of them writes
signal(protect_check);
// READ
wait(protect_check); // CS to change and check rc
rc--;
if rc == 0 then signal(noone_accesses) fi // "last" reader: signals
signal(protect_check)
forever
```

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Deadlock avoidance using the OS as arbitrator

Resource request & allocation is managed by the OS (note: we are talking of resource-request admission, not CPU scheduling!)

Deadlock avoidance:

- deadlock *might be* possible if resources are granted arbitrarily,
- but OS uses extra info to grant requests and schedule processes so that it avoids deadlock
- **Banker's algorithm [Dijkstra, 1965]** can be used to decide if a resource request can be granted safely

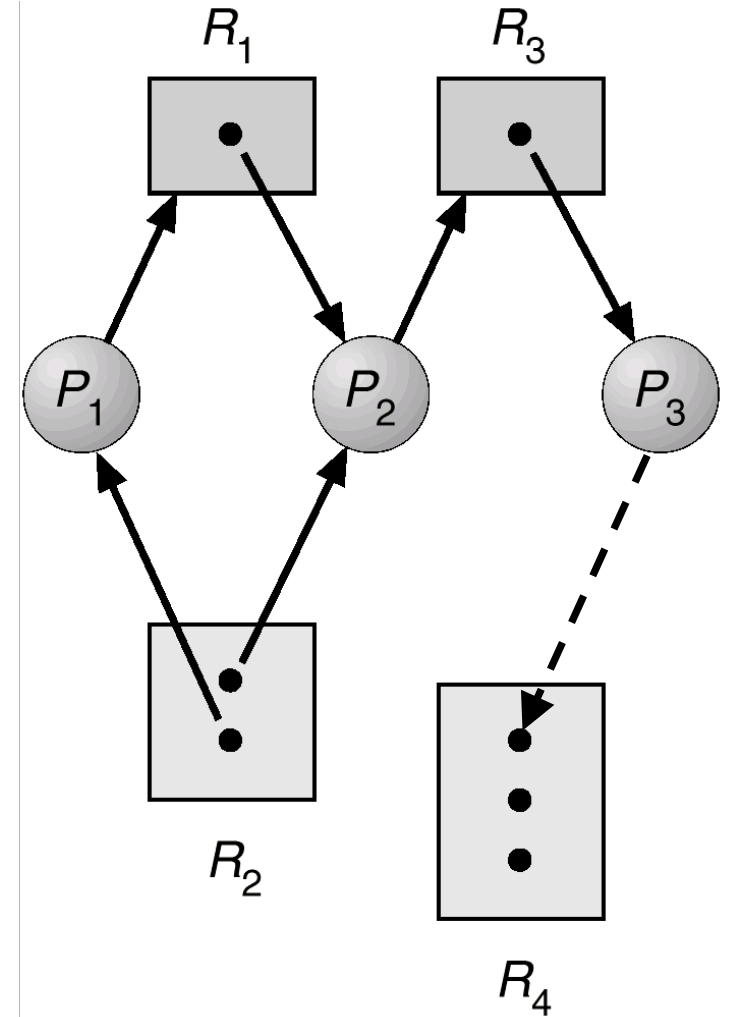
Deadlock avoidance by the OS: system model

Resource types $R_1, R_2, \dots, R_m$

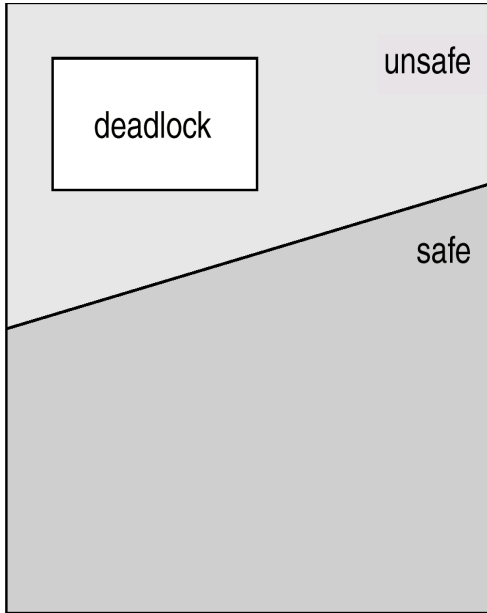
- e.g., I/O devices, files
- each resource type R_i has W_i instances.

Resource-Allocation Bipartite Graph $G(V,E)$

- nodes:
 - $P = \{P_1, P_2, \dots, P_n\}$ the set of **processes**
 - $R = \{R_1, R_2, \dots, R_m\}$ the set of **resource types**
- edges:
 - **claim edge**: $P_i \rightarrow R_j$
 - If dotted: potential request
 - **assignment edge**: $R_j \rightarrow P_i$



Deadlock avoidance using the OS as arbitrator

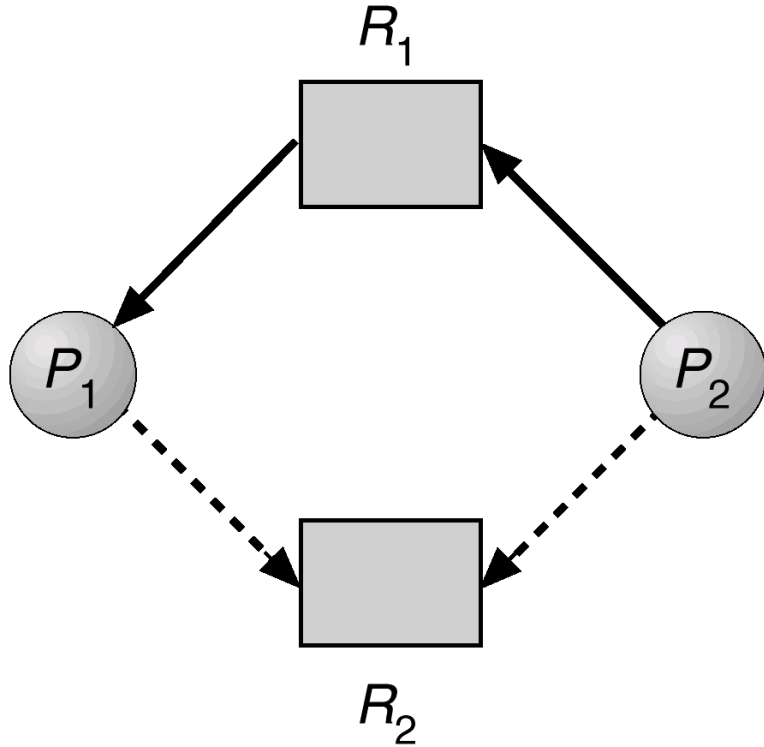


Deadlock-avoidance algorithm, run by the OS:

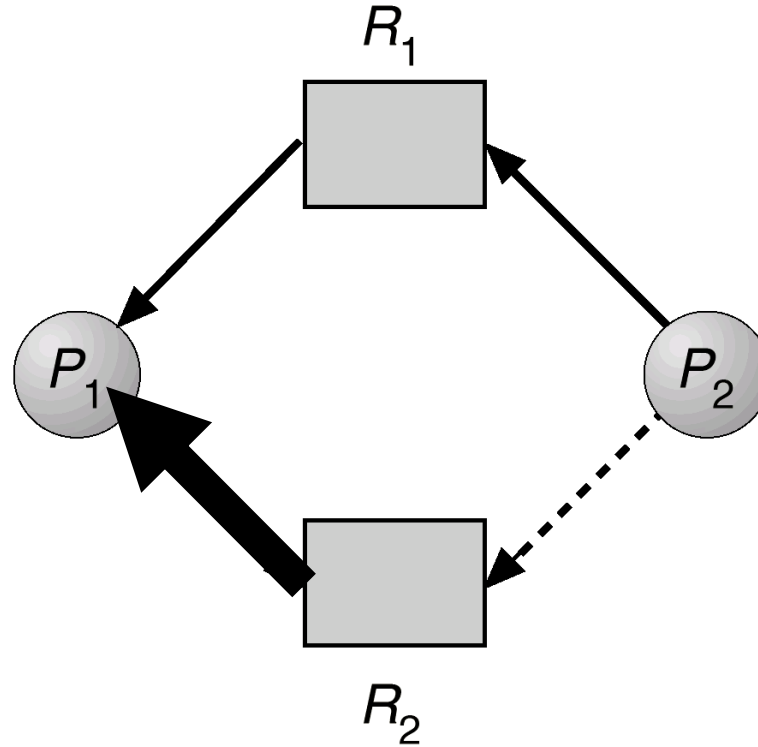
- examines the **resource-allocation state**...
 - *Available, allocated resources*
 - *maximum possible demands* of the processes.
- ...to **ensure there is no potential deadlock**:
 - **unsafe** state $\Rightarrow$ **deadlock might** occur (i.e., later, if all processes request their maximum and no one can be granted)
- **Avoidance** = ensure that system will not enter an *unsafe* state, by **delaying processes with risky requests**, until enough resources are freed.



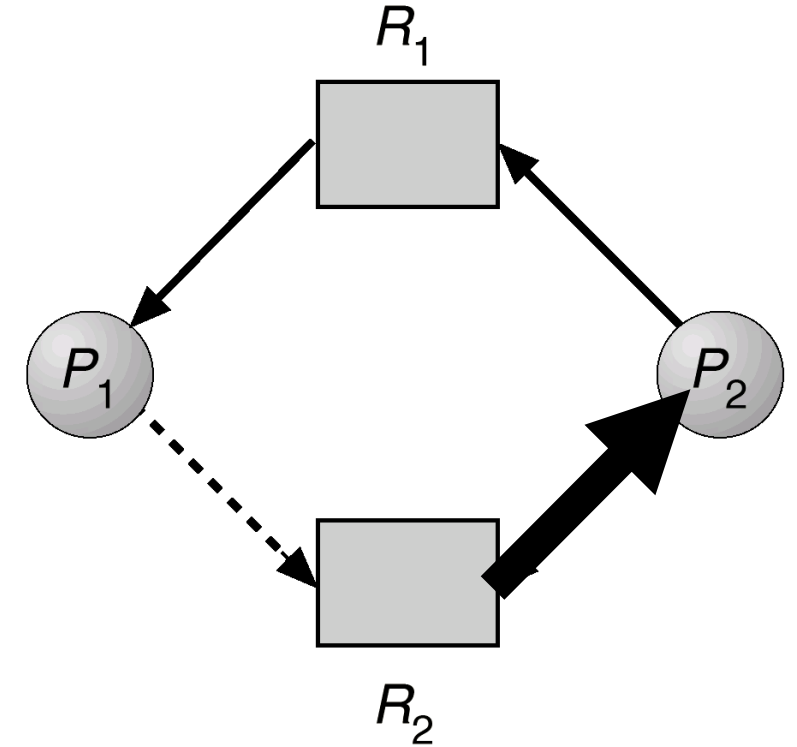
Deadlock avoidance by the OS: example



Example Safe State



Safe State



Example Unsafe State:
i.e., granting the P2-R2 request should not be made until P1 finishes (but P2 could run, the key point here is not granting R2 to P2)

Deadlock avoidance by the OS: banker's algorithm



Max [i,j] = k:

P_i may request max k instances of R_j .

Allocation[i,j] = k:

P_i holds k instances of R_j

Available [j] = k:

k instances of R_j are still available.

Avoidance = ensure that system will not enter an unsafe state.

Idea:

If *potentially satisfying a request can* result in an *unsafe state* // i.e., the bank will have not enough money to let its customers finish and return their loans in case someone requests its max needs

then the *requesting process is suspended*

until enough resources are freed // by processes that where holding them

How to do the safety check efficiently?

Banker's algo gives criterion that can be checked using the Max, Allocation, Available matrices

Deadlock detection and recovery

- If OS grants requests without checking safety upon every request
- It can allow the system to enter a deadlocked state and when detected (e.g., through detection of cyclical waits when each resource type has one instance), *recover*

- (1) **Process Termination:** Abort all or some deadlocked processes until the deadlock is eliminated.
- (2) **Resource Preemption:** Select victim and rollback – return to some safe state, restart process from that state (only for suitable resources and processes)

Must decide on a selection criteria (cost, starvation risks, ...)

Recovery is pretty expensive as a method